

Simran Koul

PhD Student
Bren School of Environmental Science & Management
University of California, Santa Barbara

Email: [Simran Koul](mailto:Simran.Koul@ucsb.edu)

Profile links:  
  

RESEARCH INTEREST: Air Pollution and Reproductive Health | Climate-Related Environmental Stressors | Population Health Inequality

EDUCATION

- PhD in Environmental Science and Management Sept, 2024-date
University of California, Santa Barbara.
- Master of Science in Environment Management (*Gold Medalist*) Oct, 2020-Jul, 2022
Guru Gobind Singh Indraprastha University, India.
- Bachelor of Science in Botany Jul, 2017-Sept, 2020
University of Delhi, India.

FELLOWSHIPS/SCHOLARSHIPS

- Schmidt Family Foundation Research Accelerator award (**\$9000**) 2026
- Office of International Students and Scholars (OISS) Students in STEM Scholarship, UC Santa Barbara (**\$2,000**) 2026
- Bren Environmental Leadership (BEL) Fellowship, UC Santa Barbara (**\$8,500**) 2026
- Aspen Environmental Profession Student Scholarship (**\$1,000**) 2025

RESEARCH EXPERIENCE

- Graduate Student Researcher | Bren School of Environmental Science & Management | UC Santa Barbara Apr, 2026-Jun, 2026
Supervisor: Prof. Sarah Anderson
Project: Spatial AI for Wildfire Risk - Defensible Space Detection from Satellite Imagery
- Graduate Student Researcher | Bren School of Environmental Science & Management | UC Santa Barbara Oct, 2024-Jun, 2025
Supervisor: Prof. Jayajit Chakraborty
Project: Air Pollution and Social Deprivation in India
- Dissertation Intern at Indian Institute of Technology, New Delhi Jan, 2022-Jul, 2022
Supervisor: Prof. Sagnik Dey
Project: Disparities in Air Pollution Exposure and Disease Burden in Rural India
- Intern at Indian Institute of Tropical Meteorology, New Delhi. Jul, 2021-Sep, 2021
Supervisor: Dr. Deewan Singh Bisht
Project: Air Quality Assessment at an Urban Site in Delhi during 2016

BOOK CHAPTERS

1. Chakraborty, J., & **Koul, S.** (2026). Environmental Injustice. In G. Bennett & E. Goodall (Eds.), *The Palgrave Encyclopedia of Disability* (pp. 1–8). Springer Nature Switzerland.
https://doi.org/10.1007/978-3-031-40858-8_496-1
2. Tripathy, P., Agrawal, A., & **Koul, S.** (2025). Cloud-based flood mapping with Sentinel-1 SAR on Google Earth Engine: Insights from the global flood mapper. In *Data-Driven Earth*

Observation for Disaster Management: From Theory to Practical Applications (pp. 609-622). Elsevier. <https://doi.org/10.1016/B978-0-443-33803-8.00045-7>

GRANTS

- I-GUIDE Summer School Travel Award, National Science Foundation to attend the [I-GUIDE Summer School 2026: Spatial AI and Convergence Science](#), University of Illinois Urbana-Champaign, Urbana, IL, July 13–17, 2026.
- Bren School Conference Travel Grant, UC Santa Barbara (\$500) to present at the AAG Annual Meeting, San Francisco, CA, March 17–21, 2026.
- Graduate Student Association Travel Grant, UC Santa Barbara (\$400) to attend the AGU Annual meeting in San Francisco, 15–19 December 2025.
- UC Santa Barbara Technology Subsidy Grant (\$500) for academic support, awarded in 2024.

HONORS & AWARDS

- [Jeanne X. Kasperson Student Paper Award](#), Hazards, Risk, and Disaster Specialty Group, Association of American Geographers (AAG) **2026**

ACADEMIC CONFERENCES

1. **Koul, S.**, & Baylis, K. (2026, May 15). Air Pollution & Timing of Menarche in India. *Mini Conference on Women's Health, Healthy Communities Project*, University of California, Santa Barbara, CA.
2. **Koul, S.**, & Baylis, K. (2026, April 3). Early-life air pollution exposure and timing of menarche. Selected oral presentation, *2026 Collaborative Research Symposium*, Cottage Health Research Institute, Santa Barbara, CA.
3. **Koul, S.**, & Chakraborty, J. (2026). Unequal risks: Coal mining hazards and social vulnerability in India. *AAG Annual Meeting 2026*, 17-21 Mar 2026, Jeanne X. Kasperson Student Paper Award Session, <https://aag-meetings.secure-platform.com/aag2026/solicitations/93/sessiongallery/25444/application/48347>
4. **Koul, S.**, & Chakraborty, J. (2025). Exploring the Relationship between Air Pollution and Social Deprivation in India. *AGU Annual Meeting 2025*, 15–19 Dec 2025, <https://agu.confex.com/agu/agu25/meetingapp.cgi/Paper/2003005>

PROFESSIONAL POSITIONS

- [Graduate Student Associate](#), **Broom Center for Demography**, UC Santa Barbara **2026 – present**

COURSES / WORKSHOPS ATTENDED

- [I-GUIDE Summer School 2026: Spatial AI and Convergence Science](#), University of Illinois Urbana-Champaign, Urbana, IL **July 13–17, 2026**
- [Environmental Data Science \(EDS\) Summit 2026](#), National Center for Ecological Analysis and Synthesis (NCEAS), Santa Barbara, CA **January 27–29, 2026**

JOURNALS REVIEWS

[Theoretical and Applied Climatology](#)

[Global Environmental Politics](#)

TEACHING

Courses taught at UC Santa Barbara

ESM 205	<i>Environmental Justice. Winter 2026 (Graduate)</i>
SOC 1	<i>Introduction to Sociology. Fall 2025 (Undergraduate)</i>
INT 95A	<i>Modern Research University. Summer 2025 (Undergraduate)</i>

INT 93LS

Research in STEM (Policy Puzzle). **Summer 2025 (Undergraduate)**

MENTORING

- Anoushka Rajeshkumar (High School) **Summer 2026**
- Maria Traslado (Undergraduate, BEL Fellow) **Summer 2026**
- Juliana Woehl (Undergraduate) **Fall 2025, Winter 2026**

COMMUNITY SERVICE

- Monthly Space Dinner Volunteer at [The Transition House](#), Santa Barbara, CA. **2025-date**
- Notetaker, Santa Barbara [Wildfire Resilience Collaborative](#) - Inaugural Symposium, Santa Barbara, CA. **2026**

REFERENCE

- **Kathy Baylis**, Professor | Department of Geography | UC Santa Barbara | baylis@ucsb.edu
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